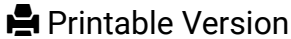


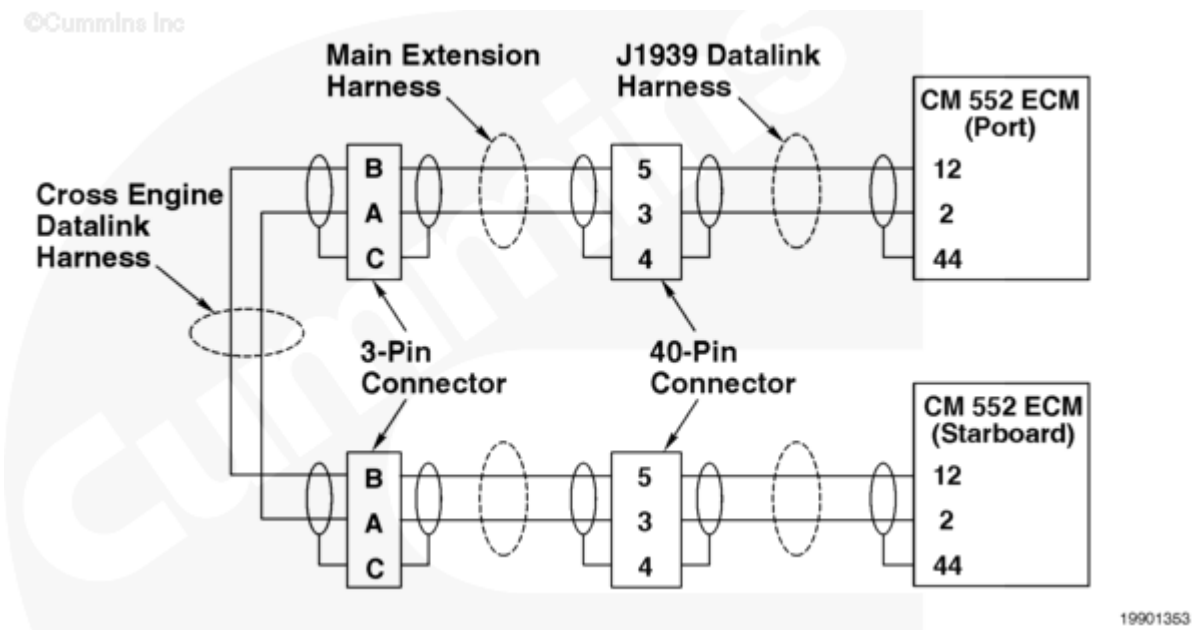
# Fault Code: 237

## Multiple Unit Synchronization Error



# Overview

| Codes                                                                                                  | Reason                                                                                                  | Effect                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Fault Code: 237</p> <p>PID(P): S30</p> <p>SPN: 644</p> <p>FMI: 2</p> <p>Lamp: Amber</p> <p>SRT:</p> | <p>External speed input (multiple unit synchronization) - data erratic, intermittent, or incorrect.</p> | <p>Engines will <b>not</b> be able to be controlled with one throttle input. The secondary engine <b>must</b> be controlled separately via its throttle input.</p> |



## Multiple Unit Synchronization Circuit

## Circuit Description

The primary engine outputs a throttle signal to the secondary engine via the J1939 interconnect. This is activated by a switch closure. This signal is interpreted by the secondary engine and is used as the throttle signal input by the secondary engine.

## Component Location

**Not** applicable.

## Shoptalk

The secondary engine expects a throttle signal to be transmitted from the primary engine via the J1939 datalink. A faulty signal will generate Fault Code 237 in the secondary engine.

Fault Code 237 can occur when the secondary keyswitch is ON and the primary keyswitch is OFF. It can also occur when there is an open circuit in the wires that connect the J1939 signal from the primary to the secondary engine. The diagram above illustrates the circuit and the data flow path.

- Verify the multiple unit synchronization jumpers are in place.
- Verify there is **only** one primary jumper and one secondary jumper.
- Verify both keyswitches are ON.
- Toggle the synchronization switch to determine if the fault goes active and inactive.

Verify that the SAE J1939 datalink is active and communicates correctly with INSITE™ electronic service tool on all engines. Refer to Procedure If any ECM will not communicate over the SAE J1939 datalink, follow Procedure 022-999 (Service Tools and Hardware - Overview) in Section F. (</qs3/pubsys2/xml/en/procedures/00/00-022-999.html>)

Refer to Troubleshooting Fault Code t05-237 (</qs3/pubsys2/xml/en/procedures/07/07-t05-237.html>)